Quantifying the Impact of Deque University on Real-World Accessibility Outcomes

DS Capstone Project Description Document

Name of Project:

Quantifying the Impact of Deque University on Real-World Accessibility Outcomes

Company Name:

Deque Systems

Mentor Name and Email:

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Part 1: The Project Description

Ideas and Goals:

Deque University is the world's leading curriculum for learning digital accessibility. Hundreds of thousands of learners have taken Deque University courses over the twelve years since Deque University was first created. What kind of impact, if any, has Deque University had on the most important metric of all: the accessibility of digital content? If Deque University students don't go on to make the world more accessible, then it doesn't matter how many courses they took, or how highly the courses were rated. In the end, the only thing that matters is whether Deque University has a measurable impact on real-world accessibility outcomes. This project will investigate the factors that do or don't lead to accessible outcomes.

Applications:

There are two main categories of Deque University customers: 1) enterprise accounts with multiple learners (sometimes in the tens or hundreds of thousands) and 2) individual users with personal subscriptions just for themselves. Enterprise account admins seek assurance that a Deque University subscription offers a worthwhile value proposition in achieving accessible websites, mobile apps, and other digital content. Individuals also value outcomes, though not all individuals oversee websites or mobile apps, so we will investigate other outcomes as well, such as the exam pass rate for professional certification in accessibility, and career advancement. The hypothesis is that Deque University offers clear value propositions for both categories of customers, but we'll follow the data where they lead us.

Structure:

Part 1: Research Design

Most, if not all, of this phase of the project should be complete before the beginning of the student involvement, but we plan to give the students the opportunity to review and critique the research design, including the planned data analyses.

Part 2: Review of past literature and studies

Deque staff will complete an initial literature review, at least in raw format. Students will continue this task, reviewing, critiquing, and gleaning insights from other published studies, literature, and data sets.

Part 3: Identify Suitable Comparisons

The Deque team will identify suitable comparisons (e.g. other organizations analogous to the Deque clients in this data analysis), to allow for a meaningful discussion of outcome metrics.

Part 3: Data Collection and Cleaning

If there are any data remaining to be collected, the student data science team will assist in this task. The team will also help clean the data to bring the data sets to a state suitable for analysis.

Part 4: Data Analysis

The student data science team will analyze the data according to the research design.

Part 5: Executive Summary

The student data science team will prepare a report on the outcomes.

Challenges:

In the research design, the researchers (primarily Deque employees at this stage) will need to establish working relationships with clients willing to participate in the research, and define which users are responsible for which aspects of a company's digital presence. Challenges may arise in navigating the customer relationship, and in the data gathering. In the data analysis phase, cleaning and preparing the data will take time and care.

Expectations:

Students are expected to commit 8-10 hours per week, engaging actively in team meetings and contributing to the project's development. The project does not require U.S. citizenship for participation. Deliverables include the collection and analysis of the data, and a presentation analyzing the data.

Team Dynamics and Leadership:

Teams will self-organize with elected leaders, and mentors will provide medium-level involvement, offering weekly feedback and support. Communication tools like Zoom, Discord, Slack, or WhatsApp will be used for both synchronous and asynchronous interactions.

Student Background:

- A working knowledge of HTML would be helpful.
- Team members should have a desire to make the world more inclusive for people with disabilities.
- A working knowledge of JavaScript is *NOT* required, but could help, because some of the learning content involves JavaScript code, logic, or conventions.

"Given" Elements:

Team members will be given access to

1) Deque University (a full curriculum of courses about accessibility, including IAAP certification preparation courses)
2) Deque's axe Assistant (an AI chat tool)
3) Khan Academy's AI Tutor, Khanmigo
4) Deque University usage data sets (access logs, feedback forms submitted, Google Analytics, etc.)
5) Access to Deque's enterprise web accessibility software tools for finding and correcting web accessibility errors

Deliverables:

1) Initial comments and critique of the research plan
2) An expanded literature review, building on the sources provided by Deque
3) Data sets that have been cleaned, organized, and prepared for analysis
4) A presentation of the analysis of the data, for the executives and managers at Deque.

Section 2 Student and Team Options

Section 1:

1) **Number of Teams you prefer for this Project:**
    a. One team
2) **Number of students in a Team:**
    a. 2-7
3) **Meetings:**
    a. **Project kick-off:** All Mentors are asked to meet with their Team(s) via Zoom around Week 3 (~ Feb 2 – Feb 9) to "kick-off the project.
    b. **Synchronous meetings per month:** Ideally, we would like to meet with the team once per week, to establish a regular cadence to the work, and to discuss ideas as they evolve, but we will allow some flexibility to accommodate schedules.
    c. **Asynchronous meetings:** We can utilize any
2) **What "style" of leadership do you expect from your team(s)?**
    i. **Medium:** Teams will elect their own Team Leader and will mostly but not fully lead, run, and manage the project. Mentor participation will be "medium", such as weekly feedback, answering questions, medium direction. In this case, there are some specific steps and expectations. The Team AND the Mentor together will make the Project design and direction decisions.
3) **What is the nature of the project?**
    a. The team will analyze the effectiveness of Deque University in achieving more accessible outcomes in web and digital content, when compared to comparable groups that do not use Deque University
4) **Do students need to be US Citizens to participate in the project?**
    a. No.
5) **What backgrounds or prerequisites would you prefer that students have?**
    a. A working knowledge of HTML would be helpful.
    b. Team members should have a desire to make the world more inclusive for people with disabilities.
6) **How many hours per week do you expect from each student on your team?** 8-10
7) **Links or data or websites that might help students to better understand the overall nature of the focus of the project:**
    a. Dequeuniversity.com
    b. Deque.com
    c. Accessibilityassociation.org
    d. https://www.khanacademy.org/settings/khan-labs
8) **In one paragraph, please describe the expectations you have of your student team with respect to the project, deliverables, meetings, engagement, etc.**
    a. Deque's mission is digital equality, meaning that our goal is to make the world's digital content accessible to people with disabilities. Deque produces tools and training to this end. The question is: does Deque have a positive impact? Or is there no significant

difference between outcomes whether Deque is involved or not? Specifically, do the courses in Deque University lead to more accessible web sites, documents, and mobile apps? If the answer is no, then Deque University is not accomplishing its mission, and we want to identify and quantify the factors leading to this failure. If the answer is yes, we want to highlight how, and under what circumstances, Deque University is being leveraged to accomplish accessible outcomes.